Biospectra : Vol.9(1), March, 2014. pp 187-192
An International Biannual Refereed Journal of Life Sciences

Plant Sciences

ISSN : 0973-7057

Callus induction and regeneration of local varieties of rice (*Oryza sativa L.*)

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Received , 24th November, 2013 ;Revised: 15th January, 2014

Abstract : There are large numbers of local varieties in the state of Jharkhand which grow readily without much input and carry many important traits. Despite superior qualities these varieties are under sheer negligence due to introduction of high yielding varieties. It is apprehended that our valuable local varieties will be lost for ever if some measures are not undertaken timely. Therefore, there is an urgent need to protect and to encourage the cultivation of local varieties of rice amongst the local farmers before it is too late. In these perspectives biotechnology could be instrumental in qualitative and quantitative improvement of local rice varieties as tradition plant breeding has its own limitations. In the present study four local varieties; Goda Dhan, Pasara, Senegal Baba and Karhani have been taken which are purely local varieties grown and consumed by ethnic communities residing in the Jharkhand state. These varieties have been put in cultural conditions to screen their callus induction and regeneration potentials to screen their morphogenic potentials, so that they may be identified for rice improvement programme through genetic manipulation. The variety Karhani proved to be best variety in terms of callus induction and regeneration frequencies from seed accounting 55and 90%, respectively. Other varieties; Senegal Baba (70% and 85%), Pasara (50 and 60%) and Goda Dhan (55 and 63%) show decreasing orders for callusing and regeneration. The responses are based on genotype of rice varieties.

Key Words: Karhani, Senegal Baba, Pasaras, Goda Dhan, callusing, Regeneration.

INTRODUCTION

Rice (*Oryza sativa* L.) is the second largest food crop consumed after wheat worldwide. In view of the population explosion both qualitative and quantitative increase in food crops is inevitable. However, the conventional method of plant breedi ng is applied for crop improvement but it has own limitations. Therefore, there is an urgent need to find out an alternate strategies for quantitative and qualitative improvements so as to meet the food demands. In these perspectives tissue culture technique holds much promise, therefore, can be utilized as an effective tool of biotechnology for fruitful results.

Jharkhand is a tribal dominated region, which is very rich centre of biodiversity in rice crop plants, their wild relatives/ land races. These local varieties are carrying many important traits; such as short duration maturity, grow easily on inhospitable conditions without much input, abiotic and biotic stress tolerant. Therefore, there is an urgent need to protect and improve these varieties through tissue culture so that environment and soil health may be maintained and cultivation of local varieties may be the choice and become common practice amongst the farmers. Although rice tissue culture has been successfully established all over the world but still there are huge numbers which need to be explored for morphogenic studies as cultural response vary greatly with genotypes, organs of the plants and developmental stages of the organs. Other parameter, for e.g. culture medium is also of no less importance. Moreover, role of tissue culture in artificial induction of genetical variables in the rice and other cereal improvement programme has been fairly proved over the decades by large numbers of workers.[1,2,3,4] Therefore, before any meaningful work, it is imperative to establish

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profuse callus induction and subsequent regeneration of plant.[5-14]

With this view four local varieties have been identified to bring them under screening process for their morphogenic potential for any advance studies.

MATERIAL AND METHOD

In the present investigation seeds of four local varieties of rice (*Oryza sativa* L.); Goda Dhan,Pasara, Senegal Baba and Karhani were collected from local farmers and Birsa Agriculture University, Kanke, Ranchi. Identified and selected varieties show some important features;

1. Goda Dhan: An upland variety; maturity 70days, panicle length- 25cm; husk colour- straw yellow; grain colour- red and resistant to bacterial leaf blight.

2. Pasara : An upland variety sown by broadcasting; maturity 90-100 days; panicle length 23 cm; drought tolerant; resistant to bacterial leaf blight, No. of grains per panicle 100-110; husk colour- straw white and grain colour- white.

3. Senegal Baba: A midland variety sown by broadcasting; maturity-110 days; panicle length: 26-30cm; no. of grains per panicle: 140, husk colour: straw white, grain colour- red.

4. Karhani : A midland variety sown by broadcasting; maturity- 123 days; panicle length: 26.5 cm; no. of grains per panicle-170; husk colour: white; grain colour-red.

Experimentation was carried with the seed explants of different varieties. In order to obtain sterile explants, seeds were dehusked manually, dipped in 70% ethanol for 1 min followed by treatment of 0.01% mercuric chloride solution for 15 to 20 min and finally rinsed 3 times with sterile distilled water .

The sterile materials were inoculated on MS (Murashige and Skoog, 1962) supplemented with 2.0 mg/L 2, 4-D. The pH of the medium was adjusted to 5.8 with acid (N/10HCl) or alkali (N/10 NaOH). The rice seeds, 2-3 in numbers were inoculated into each culture tube under a laminar airflow cabinet.

Regeneration efficacy was observed with MS medium supplemented with 2 mg/L IAA + 4 mg/L KN. All cultures were incubated at 25 $\pm$ 3°C with a 16 h light and 8 h dark at 2000 lux light intensity with 60% relative humidity. For callus induction the cultures were initially placed under dark condition for 10-15 days. Weekly visual observation of cultures was made and frequency of callus induction and regenerations were recorded.

RESULTS AND DISCUSSION

In seed culture the hypertrophy of tissues was observed in 4-6 days and full growth of callus were attained within 4-6 weeks in MS+2,4-D (2mg/L). The nature of callus, rate of growth and colours were dependent on the genotype up to certain stage only, thereafter marked differences were completely lost. The callus induction frequency from the seed was maximum in Karhani (80%), followed by Senegal Baba (70%) and Goda Dhan (55%)and Pasara(50%), (Table 1; Fig 1). In seeds, calli were derived from scutellum showing either profuse mass of callus or sometimes coleoptile emerged together with the callus (Fig.2A,B). In the present investigation percentage of callus formation was genotype dependent. These observation is in conformity with works of earlier workers.[5,,14-18,]

Calli derived from different explants when transferred to differentiating medium, MS+ IAA (2mg/L) + KN (4mg/L)or simple plain medium. Differentiation was initiated within a week with notably enlargement of the callus followed by the appearance of green spots which later turned into shoots. Root initiated late and thus complete plantlets were regenerated. In addition to whole plants other responses were also noticed, such as shoots only, roots only, unresponsive or necrotic callus (Fig.1 C, D). The whole plant was regenerated within 4-6 weeks (Fig. 2E). The regeneration depends upon the nature of callus. The embryogenic callus in the present investigation, is generally dry, compact and nodular, which show embryogenesis as the main pathway of regeneration in all the four cultivars. These observations are in agreement with the findings of other workers.[19,20,21,] Variety Karhani showed 90% regeneration into whole plant while Senegal Baba, Goda Dhan and Pasra the best response of 85, 63 and 60%, respectively. Genotypes and types of explants were the major factors on culture response. In this experiment, there was significant difference among different genotypes. The variance of different genotypes resulted from the different

physiological characteristics and developmental stages, which led to the different concentrations and ratios of endogenous hormones.[22,] In the present investigation genotypes had great bearing on morphogenic response. It accorded with the result of earlier workers.[23-29]

Out of the four varieties Karhani, which is a pure local variety proved to be the best cultivars and therefore, could be identified for higher studies through genetic improvement programmes.

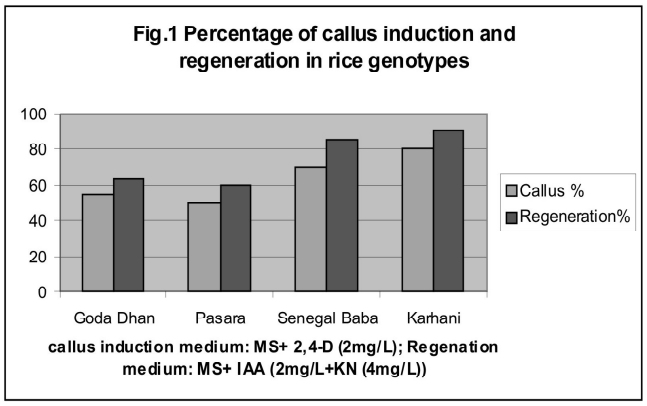
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Table 1: Effect of genotype on callusing and regeneration percentage

Genotype	No. of Seeds	Degree of callusing on MS+ 2,4-D (2.5mg/L)	callus %	Regeneration % MS+IAA (2mg/L)+KN (4mg/L)			
				NR	Roots	shoot	Root + Shoot
Goda Dhan	40	+++	55	05	20	12	63
Pasara	40	+++	50	07	18	15	60
Senegal Baba	40	+++	70	00	05	10	85
Karhani	40	+++	80	00	02	08	90

(+): good (++): Very good (+++): Excellent; NR: No Response.



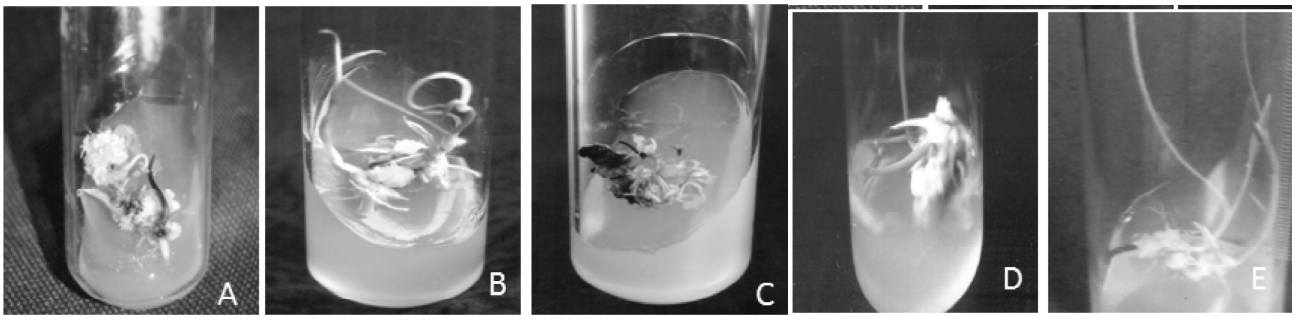


Fig.1 Callus induction and regenerationof local varieties of rice (*Oryza sativa*L.): A- callus + small Coleoptile; B- Callus+ enlarged coleoptile; C- Callus showing roots and necrotic regions; D- Regeneration of shoot and E- Multiple shoots and roots.

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